Specification Tests

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In a regression model we may want to test whether our model specification is correct:

$$\begin{aligned} & H_0 \,:\, E(Y \mid X) = m(X, \theta) \quad a.s. - X \quad \text{for some} \quad \theta \in \Theta \\ \text{against} \quad & H_1 \,:\, P(E(Y \mid X) = m(X, \theta)) < 1 \quad \text{for all} \quad \theta \end{aligned}$$

The idea is always to compare the difference of some quantity(residual, squared residual, likelihood, variance) under the null and the alternative.

The history seems to start with Pearson's χ^2 test, according to Wiki, the time series Ljung-BOx test for auto-correlation and likelihood ratio statistics are cases of χ^2 tests. I didn't read the paper of Pearson, but it is refered to in Andrews specification test using sample splitting. But the literature seems not to like the sample-splitting method.

1 Parametric Alternative

Hausman, 1978 tests parametric null against parametric alternative and gives rise to the specification test literature. By comparing an efficient and non-efficient estimator. White has a paper on nonlinear regression.

W. Newey, 1985 and W. K. Newey, 1985 are two articles that summarize the specification test (parametric against parametric), using Maximum Likelihood and GMM respectively.

It incorporates the Hausman's test and Hansen(1982)'s test for overidentifying restrictions

2 Parametric Null Against Nonparametric Alternative

2.1 Integrated Moments Method

Bierens, 1982 and Bierens, 1990 consider ICM method. They use Fourier Transform and analyse the implication of the hypotheses. J. Li and Xiu, 2016 referenced this method.

See Y. Fan and Q. Li, 2000 for a comparison between the kernel-based method and Bierens' method(they are similar in that both use a weighted sum of squared errors, but one with kernel weights, the other with exponential).

2.2 Kernel Based Method

There is another trend where people use kernel smoothing: Hardle and Mammen, 1993 which computes the kernel smoothed $\hat{m}_{alternative}$ and compare it to $m(\hat{\theta})$ by computing the integrated squared errors.

Key component is CLT on U-statistics, see de Jong, 1987 and Hall, 1984. See also Q. Li and Racine, 2007 which I think it's more detailed. Horowitz and Spokoiny, 2001 extends the Hardle and Mammen, 1993 result by considering multiple bandwidth $h_1, \dots, h_n$ and obtain a uniformly powerful test.

2.3 Empirical Likelihood and GLR

Empirical Likelihood is a major tool. I have another note for it.

The empirical likelihood and generalized likelihood ratio statistics. The idea is to construct likelihood ratio statistic:

$$\lambda_n = -2[\log f(\hat{\theta}_{H_0}) - \log f(\hat{\theta}_{ML})]$$

but with the θ_{ML} replaced by some nonparametric estimate, usually local linear regression estimate because nonparametric MLE is hard to find and nasty. This idea is called Empirical Likelihood, due to Owen from Stanford. The EL method seems to be combined to give GEL, which is an improvement over GMM(W. K. Newey and Smith, 2004) but I haven't read it yet.

J. Fan, C. Zhang, and J. Zhang, 2001 is an important paper on this Generalised likelihood ratio test, as it proves the Wilk's pheonomenon: the nuisance parameters do not show up in the limiting distribution.

This idea of EL and GLRT gives rise to huge amount of literature. S. X. Chen, Härdle, and M. Li, 2003, etc. There is now a trend to apply this method to high-frequency data.

Because the convergence is slow, usually we use wild bootstrap to approximate the distribution in finite sample

Also, GLR statistic for dependent datat seems to be underdeveloped.

2.4 Moment Based

Chang, S. X. Chen, and X. Chen, 2015.

2.5 Based on Difference Between Distribution

Key component is the empirical process and uniform convergence result. Escanciano, Pardo-Fernández, and Keilegom, 2018 gives a test when the data is stationary and β-mixing. It requires estimation rate under both null and alternative.